Wang 2019 Glioblastoma — analysis summary

RUN 4 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

7/10 samples (≥237 cells, gap-find) on one continuous 10x manifold; per-neighbourhood iLISI 1.03→2.60/7 across θ 2–10, plateaus (θ=20 → 2.59). θ=2 is a per-patient archipelago; θ=10 intermixes into one mass. 16 cells in 1 island removed → 9,634 cells, one component.

Gate 2 — module geography → GO (non-branching)

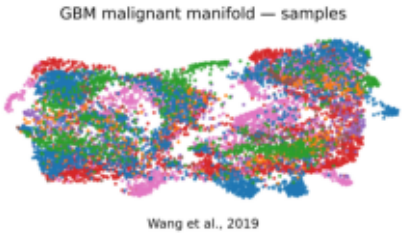
Shared Gast/pSom gradient (co-vary r=+0.16) with Crest an inverse terminal readout (Crest vs Gast r=−0.49, vs pSom r=−0.87). Crest-high decile at pseudotime 0.85 (terminal). No Gast/pSom competing arms; no Crest-on-Gast-arm inversion — no flag.

Gate 3 — hub cross-repression → NA

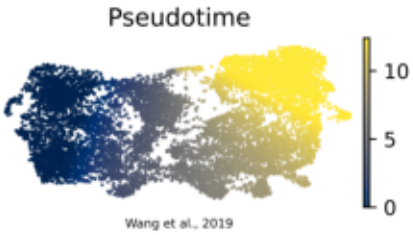
Gast/pSom co-vary (not compete) — single collapsed axis, no bifurcation to test.

STEP FIGURES

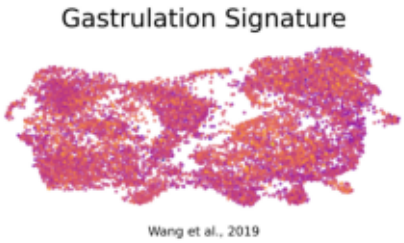
Manifold — samples



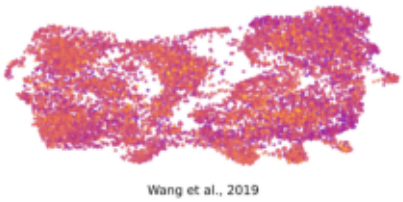
Pseudotime



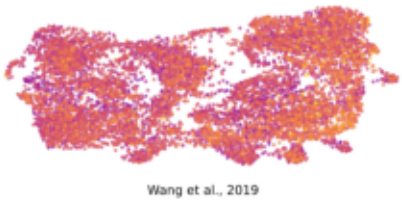
Gastrulation



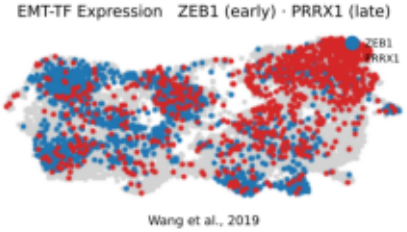
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019); 10x scRNA-seq, single leaf; 9,916 malignant raw → 9,634 final after threshold + island removal. IDH-wildtype, 7 samples.

Verdict.
Pattern reproduces as a non-branching embryonic axis: a Gast/pSom-high origin grading to a Crest-high terminus (pSom ρ=−0.61, Crest ρ=+0.48 along pseudotime), continuity 0.98 checked on an independent diffusion embedding. Linear geometry → Gate 3 NA. EMT-TFs ZEB1 (early) → PRRX1 (late). Module means Gast −0.003 / pSom +0.008 / Crest −0.007. Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min cells/sample	237 (gap 396→143, ratio 2.77; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Overdispersed genes	2,113 (scFates GAM)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Island removal	6×-median radius graph; 16 cells → 9,634	**Root node**	Gast/pSom-high, Crest-low tip
Geometry	Linear (non-branching); fold-free 2.83	**EMT-TFs**	ZEB1 (early) → PRRX1 (late)